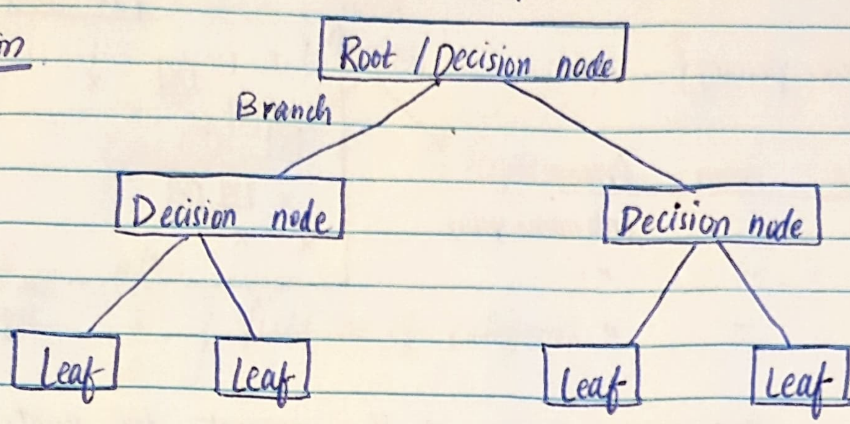
5> Decision Tree – {Classification, Regression}

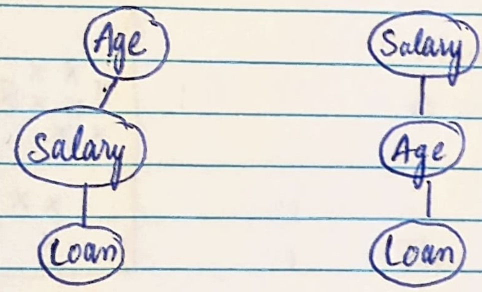
Classification



- You need to decide which root node to be selected from the decision node

Age	Salary	Loan
23	12,000	No
40	1,65,000	Yes

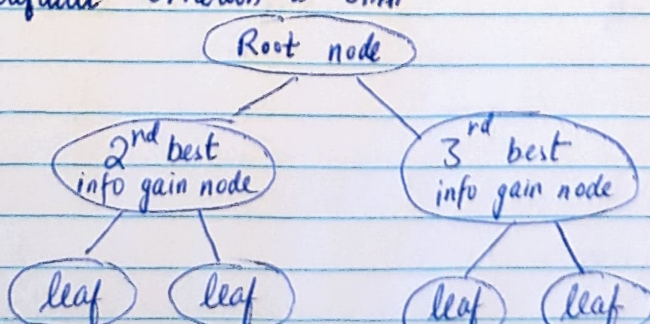
- Here what will be our root node is it age or salary



☆☆

- In order to tackle this we find the entropy of the entire dataset, then we need to calculate the information gain of each node and whoever has the highest info gain will be selected as root node.

- We use ~~entropy~~ / information gain to select the root node. mostly we use entropy. Gini is faster than Entropy. But the default criterion is Gini



Information gain is a measure of how much a feature provides info about a class.

It calculates difference between Entropy before split and average entropy after split.

$$E(S)\ \{Entropy\} = \sum_{i=1}^{c} -P_i \log_2(P_i)$$

$$\{Gini\ impurity\}\ G(S) = 1 - \sum_{i=1}^{c} (P_i)^2$$

$$\{Information\ Gain\}\ IG(S,F) = \{Entropy\}E(S) - \sum_{v \in F} \frac{|S_v|}{|S|} E(S_v)$$

Example:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	overcast	Mild	High	Strong	Yes
D13	overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

① Entropy (Dataset)

$$S = [9\ Yes, 5\ No]\quad E(S) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)$$

$$E(S) = 0.94$$

② Outlook : Sunny / overcast / Rain

S_{Sunny} [2 Yes, 3 No] $E(S_{Sunny}) = -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)$

$E(S_{Sunny}) = 0.971$

$E(S) = 0$
(pure split)

$S_{overcast}$ [4 Yes, 0 No] $E(S_{overcast}) = 0$

S_{Rain} [3 Yes, 2 No] $E(S_{Rain}) = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$

$E(S_{Rain}) = 0.971$

③ $Gain(S, outlook) = E(S) - \sum_{\gamma \in \{Sunny, overcast, rain\}} \frac{|S_\gamma|}{|S|} E(S_\gamma)$

$= 0.94 - \frac{5}{14} E_{Sunny} - \frac{4}{14} E_{overcast} - \frac{5}{14} E_{Rain}$

$= 0.94 - \frac{5}{14}(0.971) - \frac{4}{14}(0) - \frac{5}{14}(0.971)$

$G(S, outlook) = 0.24$

Similarly if we calculate the gain for temperature, humidity and wind, which ever value has gain higher that will be selected as Root Node.

Interview question

1) what is .fit()
2) what is transform()
3) what is fit_transform

Students = ['Ram', 'Sam', 'Tom']
le.fit(students)

Ram → 0
Sam → 1
Tom → 2

The encoder looks at the data and memorizes it. It does not returns output, only learns the mapping

.transform()

now the encoder already knows

Ram → 0

Sam → 1

Tom → 2

So,

le.transform(students)

returns

[0, 1, 2]

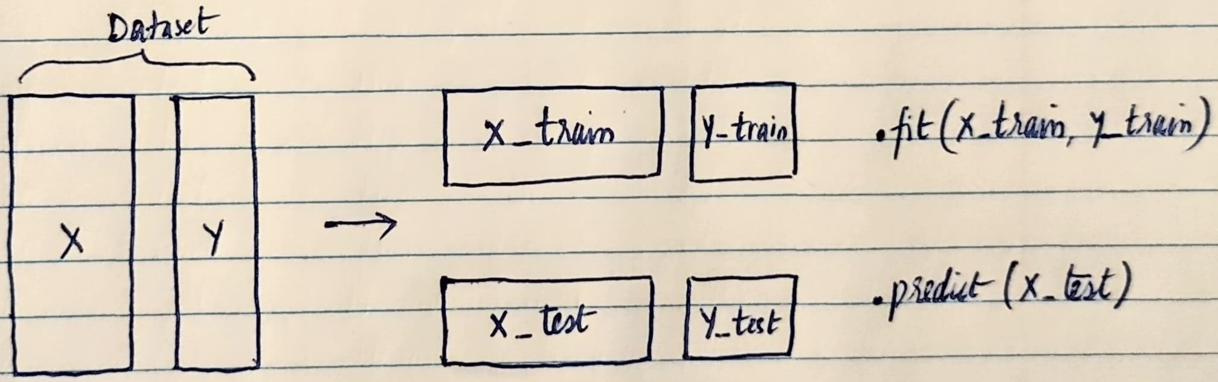
Now it converts text into numbers

.fit_transform()

it is just the combination of both .fit() & .transform()

directly returns

[0, 1, 2]



☆☆☆ Decision tree usually overfits and overconfident

Decision Tree Regressor

f_1 f_2 O/p

↓

\- \- continuous

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\-

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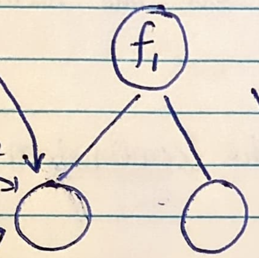
→ mean is calculated

1) f_1 feature is selected
then mean is calculated
then MSE value is computed

$$MSE = \frac{1}{2m} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

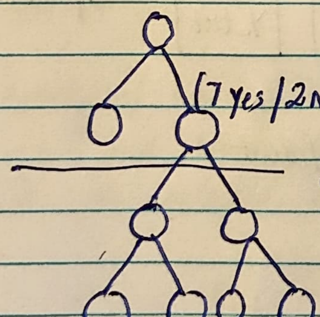
and then splitting is done

- some records go over here
- mean of these records over here
- MSE is also calculated over here



Same thing happens here, records get selected, mean and MSE gets calculated

- as the MSE gets reduced, basically we are reaching the leaf node
- MEAN is the output here
- Hyperparameters
  { pre pruning, post pruning } → to avoid overfitting



(7 yes | 2 No) → 80% chance that o/p is Yes

So here we cut the branch and this technique is called as post-pruning.

- Pre-pruning is also decided by hyperparameters called as max-depth, max_leaf. These parameters are set using gridSearchCV.